SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: Artificial Intelligence **COURSE CODE: CSA17**

COURSE OUTCOME COVERED

CO3: *Analyze knowledge representation and reasoning using propositional logic, FOL, inference, resolution, chaining methods, knowledge representation to perform logical reasoning for drawing valid conclusions and decision making. (BL4)*

Assessment Tool Weightage: 20%

Time Duration: 1 Hour
QUESTIONS: 5 **Total Marks:30**

Annexure A

Resolution Algorithm Implementation

Code of Conduct:

I ____________________ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Annexure A

Resolution Algorithm Implementation

Question 1 – Rain and Wet Ground

Knowledge Base

1. If it rains, then the ground becomes wet.
2. It is raining.

Goal

Prove that **the ground is wet** using the Resolution Algorithm.

Tasks

1. Represent the statements in propositional logic.
2. Convert all statements into CNF.

Negate the goal.

1. Apply the Resolution Algorithm step by step.
2. Show the Empty Clause (□) if the goal is proved.

Question 2 – Student Assignment Submission

Knowledge Base

1. If a student submits the assignment, then the student receives internal marks
2. Rahul submitted the assignment.

Goal

Prove that **Rahul receives internal marks** using the Resolution Algorithm.

Tasks

1. Write the propositional logic expressions.
2. Convert the premises into CNF.

Negate the goal.

1. Apply the Resolution Algorithm.
2. Conclude whether the goal is proved.

Question 3 – Library Membership

Knowledge Base

1. If a person is a library member, then the person can borrow books.
2. Priya is a library member.

Goal

1. Prove that **Priya can borrow books** using the Resolution Algorithm.

Tasks

1. Represent the knowledge base using propositional logic.
2. Convert each statement into CNF.

Negate the goal.

1. Perform resolution until no new clauses can be generated.
2. State the final conclusion.

Question 4 – Placement Eligibility

Knowledge Base

1. If a student clears the aptitude test, then the student is eligible for placement.
2. Arun cleared the aptitude test.

Goal

1. Prove that **Arun is eligible for placement** using the Resolution Algorithm.

Tasks

1. Convert the statements into propositional logic.
2. Write the CNF clauses.
3. Negate the goal.
4. Apply the Resolution Algorithm step by step.
5. Draw the resolution tree and conclude.

Question 5 – Access Control System

Knowledge Base

1. If a user enters the correct password, then the user is authenticated.
2. If a user is authenticated, then the user is granted access.
3. The user entered the correct password.

Goal

Prove that **the user is granted access** using the Resolution Algorithm.

Tasks

1. Represent the knowledge base in propositional logic.
2. Convert all implications into CNF.
3. Negate the goal.
4. Apply the Resolution Algorithm to derive the Empty Clause (□).
5. Explain each resolution step and write the final conclusion.

RUBRICS FOR CALCULATION

Numerical Problems					
Criteria	**Weightage**	**Level 4 (Exemplary)**	**Level 3 (Proficient)**	**Level 2 (Developing)**	**Level 1 (Beginning)**
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation

Question 1 – Rain and Wet Ground

1. Propositional Logic Representation

Let:

- = It is raining.
- = The ground is wet.

Knowledge Base:

1. If it rains, then the ground becomes wet.
2. It is raining.

Goal:

2. Conversion to CNF

Using:

Therefore:

CNF clauses:

-

3. Negate the Goal

Goal:

Negated goal:

Add it as a clause:

4. Resolution Steps

Resolve and :

Resolving and :

Now resolve:

Therefore:

Resolution Tree

```
    ¬R ∨ W      R
       \      /
         W
         |
        ¬W
         |
         □
```

Conclusion

Since the **Empty Clause (□)** is derived, the negation of the goal leads to a contradiction.
The ground is wet. Hence, the goal is proved using the Resolution Algorithm.

QUESTION 2 – STUDENT ASSIGNMENT SUBMISSION

Resolution Algorithm Implementation

Problem Statement

A college has a rule stating that when a student submits an assignment, the student receives internal marks. It is given that Rahul has submitted his assignment. The objective is to prove that Rahul receives internal marks using the Resolution Algorithm. This problem demonstrates how a rule and a known fact can be used to logically establish a conclusion through resolution.

Given Knowledge Base

1. If a student submits the assignment, then the student receives internal marks.
2. Rahul submitted the assignment.

Goal

Prove that Rahul receives internal marks.

1. Propositional Logic Representation

Let:

- = Rahul submitted the assignment.
- = Rahul receives internal marks.

The first statement is:
If Rahul submits the assignment, then Rahul receives internal marks.
It can be represented as:
The second statement is:
The goal is:
Therefore:
Goal:

2. Conversion into CNF

The implication:
is converted using:
Therefore:
Hence:
The given fact is:
Thus, the CNF representation of the knowledge base is:

3. Negation of the Goal

The goal is:
For resolution by contradiction, the goal is negated:
Therefore:
The complete clause set is now:

4. Resolution Process

Step 1 – Deriving Internal Marks

Take:

and:

The complementary literals are:

Applying resolution:

The literal and cancel each other.

Therefore, the resulting clause is:

Thus, the fact that Rahul submitted the assignment allows us to derive that Rahul receives internal marks.

Step 2 – Contradiction with Negated Goal

Now resolve:

with:

The complementary literals are:

Therefore:

The Empty Clause is obtained.

5. Resolution Table

Step	First Clause	Second Clause	Result
			□

6. Interpretation

The Empty Clause proves that the assumption:

is inconsistent with the knowledge base.

In other words, assuming that Rahul does not receive internal marks contradicts the facts and rules provided.

Therefore:

Hence, Rahul receives internal marks.

7.Final Conclusion

The Resolution Algorithm successfully proves the given goal. Rahul submitted the assignment, and according to the given rule, a student who submits the assignment receives internal marks.

Therefore:

The derivation of the Empty Clause confirms that the goal is logically entailed by the knowledge base.

QUESTION 3 – LIBRARY MEMBERSHIP

Resolution Algorithm Implementation

Problem Statement

A library has a rule that a person who is a library member is allowed to borrow books. It is given that Priya is a library member. The objective is to prove that Priya can borrow books using the Resolution Algorithm.

This problem illustrates the use of a conditional rule together with a known fact to derive a conclusion.

1. Propositional Logic Representation

Let:

- = Priya is a library member.
- = Priya can borrow books.

The first statement is:
The second statement is:
The goal is:
Therefore:
Goal:

2. Conversion into CNF

The implication:
can be converted into CNF using:
Therefore:
Thus, the first clause is:
The second statement is already a unit clause:
Therefore, the CNF knowledge base is:

3. Negation of the Goal

The goal is:
Negating the goal gives:
Therefore:
The complete set of clauses is:

4. Resolution Process

Step 1

Resolve:
with:
The complementary literals are and .
Therefore:

gives:
Thus, Priya's library membership allows us to derive that she can borrow books.

Step 2

Now resolve the derived clause:
with the negated goal:
The complementary literals are:
Therefore:

5. Resolution Table

Step	Clause 1	Clause 2	Resolvent
1			
2			

6. Resolution Tree

```
        ¬L ∨ B          L
           \           /
            \         /
               B
               |
               |
              ¬B
               |
               |
               □
```

7. Final Interpretation

The resolution procedure produces the Empty Clause. This means that the negated goal:
cannot be true when the knowledge base is considered.
Therefore:
This proves that Priya can borrow books.

Final Conclusion
Since Priya is a library member and the given rule states that every library member can borrow books, the Resolution Algorithm successfully establishes the desired conclusion.
The Empty Clause confirms that the goal has been logically proved.

QUESTION 4 – PLACEMENT ELIGIBILITY

Resolution Algorithm Implementation

Problem Statement

In a placement system, students are required to clear an aptitude test before becoming eligible for placement. The knowledge base states that if a student clears the aptitude test, then the student is eligible for placement. It is also given that Arun has cleared the aptitude test.
The objective is to prove that Arun is eligible for placement using the Resolution Algorithm.

1. Propositional Logic Representation
Let:

- = Arun cleared the aptitude test.
- = Arun is eligible for placement.

The first statement is:
The second statement is:
The goal is:
Therefore:
Goal:

2. CNF Conversion
The implication:
is converted using:
Therefore:
Thus:
The given fact is:
Therefore, the CNF knowledge base is:

3. Negation of the Goal
The goal is:
The negation of the goal is:
Therefore:
The complete clause set is:

4. Resolution Algorithm
Step 1 – Resolve the Aptitude Rule
Take:
and:
The complementary literals are:
After resolution:
we obtain:
Therefore, we have derived that Arun is eligible for placement.

Step 2 – Resolve with Negated Goal
The negated goal is:
The newly derived clause is:
Resolve:
The complementary literals cancel each other and produce:
Thus, the Empty Clause is obtained.

5. Resolution Table

Step	Clause 1	Clause 2	Result
1			
2			

6. Resolution Tree

```
¬A ∨ E         A
   \         /
    \       /
       E
       |
       |
      ¬E
       |
       |
       □
```

7. Interpretation of Result

The Empty Clause indicates that the negated goal:
is inconsistent with the knowledge base.
Therefore:
This means that Arun's eligibility for placement logically follows from the information given.
In practical terms, Arun has cleared the required aptitude test. According to the placement rule, clearing the aptitude test makes a student eligible for placement.

Final Conclusion

The Resolution Algorithm proves the goal successfully because the resolution process produces the Empty Clause.

QUESTION 5 – ACCESS CONTROL SYSTEM

Resolution Algorithm Implementation

Problem Statement

Consider an access control system in which a user must enter the correct password to become authenticated. Once authenticated, the user is granted access to the system. It is given that the user has entered the correct password.

The objective is to prove that the user is granted access using the Resolution Algorithm. This problem contains two implications, so the resolution process requires more than one intermediate inference before the final contradiction is obtained.

1. Propositional Logic Representation

Let:

- = The user entered the correct password.
- = The user is authenticated.
- = The user is granted access.

The first rule states:
If a user enters the correct password, then the user is authenticated.
Therefore:
The second rule states:
If a user is authenticated, then the user is granted access.
Therefore:
The third statement says:
The user entered the correct password.
Therefore:
The goal is:
Thus:

2. Conversion of Implications into CNF

First Implication

Given:
Using:
we get:
Therefore:

Second Implication

Given:
we obtain:
Therefore:

Given Fact

The user entered the correct password:
Therefore, the CNF knowledge base is:

3. Negation of the Goal

The goal is:
In resolution by refutation, the goal is negated:
Therefore:
The complete set of clauses is now:

4. Applying Resolution Step by Step

Resolution Step 1 – Authentication

Take:
and:

The complementary literals are:
Resolving these clauses gives:
Therefore:
This means that the user is authenticated because the user entered the correct password.

Resolution Step 2 – Granting Access
Now take:
and the newly derived clause:
The complementary literals are:
Applying resolution:
we obtain:
Therefore, the user is granted access.

Resolution Step 3 – Contradiction with Negated Goal
The negated goal is:
The derived clause is:
Now resolve:
The literals and are complementary.
Therefore:
The Empty Clause has been successfully derived.

5. Resolution Table

Step	Clause 1	Clause 2	Resolvent	Explanation
1				User is authenticated
2				User is granted access
3				Contradiction obtained

6. Detailed Resolution Tree

```
¬P ∨ A          P
   \           /
    \         /
        A
        |
        |
     ¬A ∨ G
        |
        |
        G
        |
       ¬G
        |
        |
        □
```

The tree shows the complete chain of reasoning.

First:
Then:
Finally:

7. Explanation of the Logical Reasoning

The first rule establishes a relationship between the password and authentication. Since the user has entered the correct password, the system can infer that the user is authenticated.

Mathematically:
The second rule establishes a relationship between authentication and access. Since authentication has been established, the system can infer that the user should be granted access.

Therefore:
However, in the Resolution Algorithm, instead of directly accepting , we add its negation:
The resolution process then derives , which directly contradicts .